MEDIUM · ADVANCED MATH · ANSWER KEY

Advanced Math

15

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 C

C. $-\frac{4}{5}$

Choice C is correct. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $5x + 4 = 0$ or $2x - 5 = 0$. Subtracting 4 from each side of the equation $5x + 4 = 0$ yields $5x = -4$. Dividing each side of this equation by 5 yields $x = -\frac{4}{5}$. Adding 5 to each side of the equation $2x - 5 = 0$ yields $2x = 5$. Dividing each side of this equation by 2 yields $x = \frac{5}{2}$. It follows that the solutions to the given equation are $-\frac{4}{5}$ and $\frac{5}{2}$. Therefore, $-\frac{4}{5}$ is a solution to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2 A

A. The height of the object 1.4 seconds after being thrown straight up in the air is 1.64 feet.

Choice A is correct. The value 1.4 is the value of x, which represents the number of seconds after the object was thrown straight up in the air. When the function h is evaluated for x = 1.4, the function has a value of 1.64, which is the height, in feet, of the object.

Choices B and C are incorrect and may result from misidentifying seconds as feet and feet as seconds. Additionally, choice C may result from incorrectly including the initial height of the object as the input x. Choice D is incorrect and may result from misidentifying height as speed.

Q3 -3

The correct answer is -3. Dividing both sides of the given equation by 38 yields $x^2 = 9$. Taking the square root of both sides of this equation yields the solutions $x = 3$ and $x = -3$. Therefore, the negative solution to the given equation is -3.

Q4 14

Accept also: −5 , −4

The correct answer is either 14, -5, or -4. The x-intercepts of a graph in the xy-plane are the points at which the graph intersects the x-axis, or when the value of y is 0. Substituting 0 for y in the given equation yields $0 = 3x - 14x + 5x + 4$. Dividing both sides of this equation by 3 yields $0 = x - 14x + 5x + 4$. Applying the zero product property to this equation yields three equations: $x - 14 = 0$, $x + 5 = 0$, and $x + 4 = 0$. Adding 14 to both sides of the equation $x - 14 = 0$ yields $x = 14$, subtracting 5 from both sides of the equation $x + 5 = 0$ yields $x = -5$, and subtracting 4 from both sides of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the x-coordinates of the x-intercepts of the graph of the given equation are 14, -5, and -4. Note that 14, -5, and -4 are examples of ways to enter a correct answer.

Q5 28

The correct answer is 28. The given absolute value equation can be rewritten as two linear equations: $4x - 4 = 112$ and $-4x - 4 = 112$, or $4x - 4 = -112$. Adding 4 to both sides of the equation $4x - 4 = 112$ results in $4x = 116$. Dividing both sides of this equation by 4 results in $x = 29$. Adding 4 to both sides of the equation $4x - 4 = -112$ results in $4x = -108$. Dividing both sides of this equation by 4 results in $x = -27$. Therefore, the two values of $x - 1$ are $29 - 1$, or 28, and $-27 - 1$, or -28. Thus, the positive value of $x - 1$ is 28.

Alternate approach: The given equation can be rewritten as $4x - 1 = 112$, which is equivalent to $4x - 1 = 112$. Dividing both sides of this equation by 4 yields $x - 1 = 28$. This equation can be rewritten as two linear equations: $x - 1 = 28$ and $-x - 1 = 28$, or $x - 1 = -28$. Therefore, the positive value of $x - 1$ is 28.

Q6

C

C. $n = 6041 + 0.004^t$

Choice C is correct. It's given that the relationship between t and n is exponential. The table shows that the value of n increases as the value of t increases. Therefore, the relationship between t and n can be represented by an increasing exponential equation of the form $n = a1 + b^t$, where a and b are positive constants. The table shows that when $t = 0$, $n = 604$. Substituting 0 for t and 604 for n in the equation $n = a1 + b^t$ yields $604 = a1 + b^0$, which is equivalent to $604 = a1$, or $604 = a$. Substituting 604 for a in the equation $n = a1 + b^t$ yields $n = 6041 + b^t$. The table also shows that when $t = 1$, $n = 606.42$. Substituting 1 for t and 606.42 for n in the equation $n = 6041 + b^t$ yields $606.42 = 6041 + b^1$, or $606.42 = 6041 + b$. Dividing both sides of this equation by 604 yields approximately $1.004 = 1 + b$. Subtracting 1 from both sides of this equation yields that the value of b is approximately 0.004. Substituting 0.004 for b in the equation $n = 6041 + b^t$ yields $n = 6041 + 0.004^t$. Therefore, of the choices, choice C best represents the relationship between t and n.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

30

Accept also: `-30`

The correct answer is either -30 or 30. Adding 7 to each side of the given equation yields $d - 30d + 30 = 0$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $d - 30 = 0$ or $d + 30 = 0$. Adding 30 to each side of the equation $d - 30 = 0$ yields $d = 30$. Subtracting 30 from each side of the equation $d + 30 = 0$ yields $d = -30$. Therefore, the solutions to the given equation are -30 and 30. Note that -30 and 30 are examples of ways to enter a correct answer.

Q8 D

D. 9,600,000

Choice D is correct. Let y represent the number of cells per milliliter x hours after the initial observation. Since the number of cells per milliliter doubles every 3 hours, the relationship between x and y can be represented by an exponential equation of the form $y = ab^{\frac{x}{k}}$, where a is the number of cells per milliliter during the initial observation and the number of cells per milliliter increases by a factor of b every k hours. It's given that there were 300,000 cells per milliliter during the initial observation. Therefore, $a = 300{,}000$. It's also given that the number of cells per milliliter doubles, or increases by a factor of 2, every 3 hours. Therefore, $b = 2$ and $k = 3$. Substituting 300,000 for a, 2 for b, and 3 for k in the equation $y = ab^{\frac{x}{k}}$ yields $y = 300{,}0002^{\frac{x}{3}}$. The number of cells per milliliter there will be 15 hours after the initial observation is the value of y in this equation when $x = 15$. Substituting 15 for x in the equation $y = 300{,}0002^{\frac{x}{3}}$ yields $y = 300{,}0002^{\frac{15}{3}}$, or $y = 300{,}0002^5$. This is equivalent to $y = 300{,}00032$, or $y = 9{,}600{,}000$. Therefore, 15 hours after the initial observation, there will be 9,600,000 cells per milliliter.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q9

A

A. $(2,3)$ and $(-2,3)$

Choice A is correct. The two equations form a system of equations, and the solutions to the system correspond to the points of intersection of the graphs. The solutions to the system can be found by substitution. Since the second equation gives y = 3, substituting 3 for y in the first equation gives $3 = x^2 - 1$. Adding 1 to both sides of the equation gives $4 = x^2$. Solving by taking the square root of both sides of the equation gives x = ±2. Since y = 3 for all values of x for the second equation, the solutions are (2, 3) and (–2, 3). Therefore, the points of intersection of the two graphs are (2, 3) and (–2, 3).

Choices B, C, and D are incorrect and may be the result of calculation errors.

Q10

B

B. $d = 6701 + 0.006^t$

Choice B is correct. It's given that the relationship between t and d is exponential. The table shows that the value of d increases as the value of t increases. Therefore, the relationship between t and d can be represented by an increasing exponential equation of the form $d = a1 + b^t$, where a and b are positive constants. The table shows that when $t = 0$, $d = 670$. Substituting 0 for t and 670 for d in the equation $d = a1 + b^t$ yields $670 = a1 + b^0$, which is equivalent to $670 = a1$, or $670 = a$. Substituting 670 for a in the equation $d = a1 + b^t$ yields $d = 6701 + b^t$. The table also shows that when $t = 1$, $d = 674.02$. Substituting 1 for t and 674.02 for d in the equation $d = 6701 + b^t$ yields $674.02 = 6701 + b^1$, or $674.02 = 6701 + b$. Dividing both sides of this equation by 670 yields $1.006 = 1 + b$. Subtracting 1 from both sides of this equation yields $b = 0.006$. Substituting 0.006 for b in the equation $d = 6701 + b^t$ yields $d = 6701 + 0.006^t$. Therefore, of the choices, choice B best represents the relationship between t and d.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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